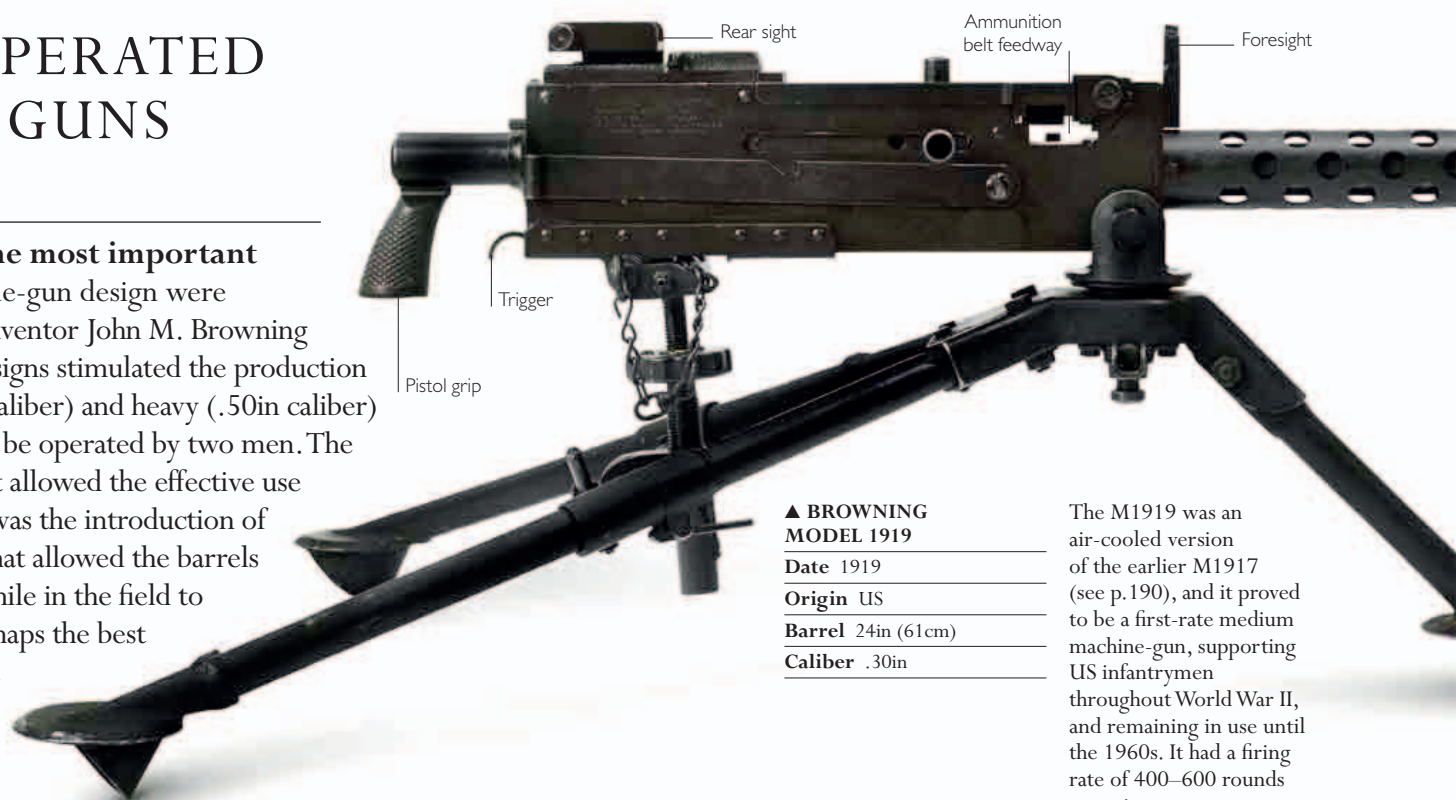


RECOIL-OPERATED MACHINE-GUNS (1918–45)

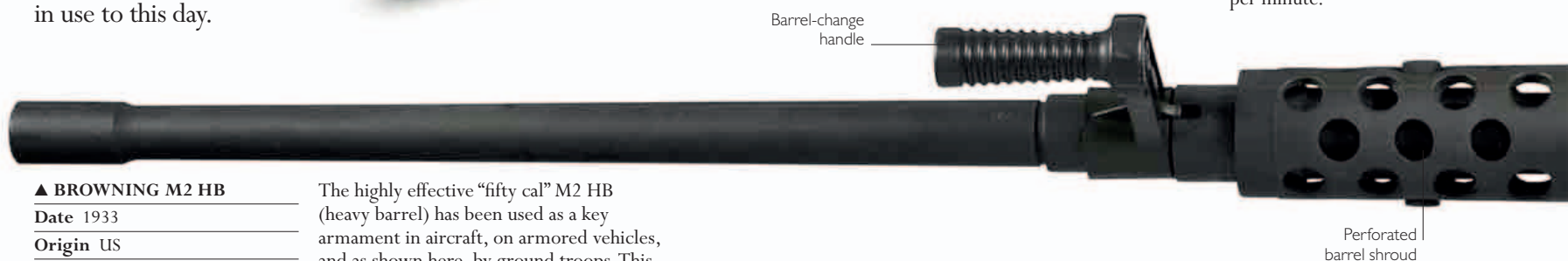
Without question, the most important advancements in machine-gun design were made by the American inventor John M. Browning (see pp. 180–81). His designs stimulated the production of both medium (.30in caliber) and heavy (.50in caliber) machine-guns that could be operated by two men. The second advancement that allowed the effective use of recoil-operated guns was the introduction of barrel-locking systems that allowed the barrels to be changed quickly while in the field to prevent overheating. Perhaps the best of these systems was that developed for use on the German MG42, a design that remains in use to this day.



▲ BROWNING MODEL 1919

Date 1919
Origin US
Barrel 24in (61cm)
Caliber .30in

The M1919 was an air-cooled version of the earlier M1917 (see p. 190), and it proved to be a first-rate medium machine-gun, supporting US infantrymen throughout World War II, and remaining in use until the 1960s. It had a firing rate of 400–600 rounds per minute.



▲ BROWNING M2 HB

Date 1933
Origin US
Barrel 3¾ft (1.14m)
Caliber .50in

The highly effective “fifty cal” M2 HB (heavy barrel) has been used as a key armament in aircraft, on armored vehicles, and as shown here, by ground troops. This gun can fire 485–635 rounds per minute, and remains in service today.

Flash hider concealed the flash of exploding propellant gases, helping to hide the firer's position



▲ MASCHINENGEWEHR 34

Date 1935
Origin Germany
Barrel 24¾in (62.7cm)
Caliber 7.92 × 57mm

The *Maschinengewehr* 34 (MG34) was a revolutionary design—light, yet robust enough to deliver sustained fire at 900 rounds per minute. However, it was difficult and expensive to manufacture and was subsequently replaced by the MG42.